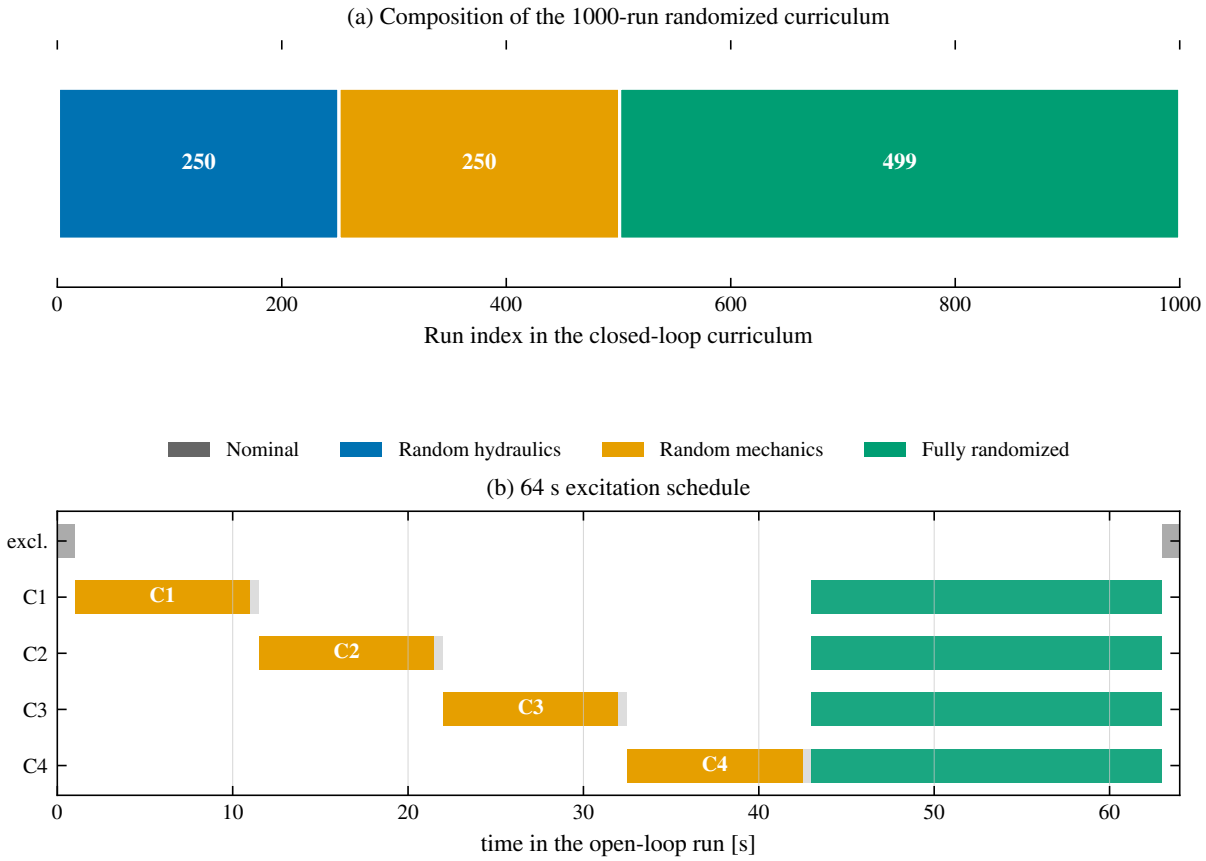


Figure 4.2: Design of the two collection campaigns. (a) Composition of the closed-loop curriculum: one nominal run, then 250 hydraulics-only, 250 mechanics-only and 499 fully randomized draws. (b) Timeline of one open-loop excitation run.

for two reasons that survive the change: it is the provenance record of the closed-loop material the actuator models were identified on, and the compliance-equivalence analysis of Section 4.2.3 is what taught this project which quantity is randomizable in a hydraulic chamber model, a lesson carried directly into the factor space of Section 4.7. Readers interested only in the models that produce results may skip to Section 4.3.

4.2.1 Sampling Design

The closed-loop campaign draws its plants from a twelve-dimensional parameter space using Latin hypercube sampling [McKay 79], which spreads the draws more evenly over each marginal than independent uniform sampling would at this sample size. The twelve coordinates are fixed once for the whole batch, before any simulation runs, so the parameter table is reproducible and can be audited independently of the data it produced.

The batch is organised as a curriculum rather than a single homogeneous pool. Run 1 is the untouched nominal plant. Runs 2–251 randomize only the hydraulics, runs 252–501 only the mechanics, and runs 502–1000 both together, giving the composition shown in Figure 4.2(a). This structure is deliberate: it makes it possible to attribute a later modelling failure to the hydraulic or the mechanical axis by retraining on one block at a time, which a uniformly mixed batch would not allow.